

1 Reflective Cognitive Runtime and Private Self-Repair

This development sequence joined six previously separate capabilities into the canonical AION cognitive runtime: persistent natural-language mission interpretation, continuous learning from independently changing outcomes, full-stack private self-repair, cross-domain neural routing, continual multi-skill composition, and metacognitive review before and after action. The resulting runtime is

$$\begin{aligned} \text{natural mission} &\rightarrow \text{governed goal} \rightarrow \text{investigate} \rightarrow \text{learn} \rightarrow \text{plan} \\ &\rightarrow \text{pre-action metacognition} \rightarrow \text{commit} \rightarrow \text{act} \rightarrow \text{observe} \\ &\rightarrow \text{criticise} \rightarrow \text{post-action reflection} \rightarrow \text{private improvement} \rightarrow \text{retain}. \end{aligned} \tag{1}$$

Neural and language components remain proposal mechanisms. HexCore governs action, externally or executably observed consequences determine success, and the Cognitive Authority Unit (CAU) remains the sole promotion authority.

1.1 Full-Stack Private Self-Repair

AION now contains one repair controller spanning perception, semantic memory, reasoning, planning, tool integration, and canonical runtime state. Given an unlabelled verified failure trace, it localises the responsible component, constructs candidate repairs in private component copies, invents forward and backward-retention checks, and promotes or rejects each component independently:

$$\begin{aligned} \text{failure trace} &\rightarrow \text{causal feature criticism} \rightarrow \text{component localisation} \rightarrow \text{private clone} \\ &\rightarrow \text{repair tournament} \rightarrow \text{forward verification} \rightarrow \text{backward retention} \\ &\rightarrow \text{self-modification audit} \rightarrow \text{CAU promotion or rollback}. \end{aligned} \tag{2}$$

The live champion remains hash-stable during experimentation. Repair priors are indexed by causal failure signature rather than by a component’s name; therefore a verified repair can transfer to a renamed failure. Ambiguous multi-component traces cause abstention instead of a forced mutation. The authority scanner rejects attempts to disable governance, modify a live component, skip forward or retention tests, erase failure history, or introduce arbitrary execution.

CAU promoted `procedure_full_stack_private_self_repair_v1`. This is governed repair of versioned component contracts, not unrestricted recursive modification of live source code or authority rules.

1.2 Persistent Natural-Language Mission Cognition

The mission interface maintains discourse state across multiple turns and process restart. It recognises research, comparison, construction, monitoring, repair, and explanation intents; resolves references to earlier entities and objectives; retains explicit corrections; and asks for clarification when the referent or intent remains ambiguous. Listener-profile state supports novice and expert explanations without altering the underlying evidence.

Resolved owner intent is compiled into a governed goal proposal containing an immutable objective hash. Interpretation does not confer execution authority:

$$\text{language interpretation} \neq \text{permission to act}. \tag{3}$$

Table 1: Full-stack private self-repair results.

Measure	Result
Cognitive component classes	6
Development repairs	6/6
Source-disjoint transfer repairs	6/6
Failure-localisation accuracy	100%
Forward repair success	100%
Backward retention	100%
Transfer attempt reduction	77.78%
Malicious self-modifications rejected	6/6
Ambiguous multi-fault abstention	Passed
Unsafe live writes	0
Terminal-objective mutations	0
Restart relearning	0

Table 2: Persistent natural-language mission results.

Measure	Result
Multi-turn sessions	10
Verified interpretations	10/10
Semantic-intent accuracy	100%
Fixed-keyword control	30%
Source-disjoint transfer	6/6
Ambiguity abstention	Passed
Correction retention	Passed
Novice and expert adaptation	Passed
Unsafe goal compilations	0
Objective mutations	0
Restart relearning	0

The first private challenger achieved nine correct sessions out of ten but misclassified an expert explanation as research. It was rejected. The failure was used to refine the private explanation representation, after which the new challenger passed all sessions without protected regression.

The promoted procedure is `procedure_persistent_natural_language_mission_interface_v1`. Its intent families and local semantic encoder remain bounded and replaceable; the result does not establish unrestricted language or social understanding.

1.3 Continuous Learning from Independently Changing Outcomes

Every committed Arena v15 consequence is converted into a canonical runtime goal. Before initiating repair, AION determines whether the observation describes a changed external world or an internal acquisition, execution, or persistence failure:

$$R(o) = \begin{cases} \text{revise working world model,} & o = \text{external change,} \\ \text{retry or rebind adapter,} & o = \text{acquisition failure,} \\ \text{private execution repair,} & o = \text{execution failure,} \\ \text{private persistence repair,} & o = \text{persistence failure.} \end{cases} \quad (4)$$

This prevents an independently changed repository, package, or weather stream from being mistaken for damage inside AION. Each retained response contains the pre-action commitment, consequence hash, authority, source revision, evidence, response class, and cognitive-cycle identifier. The service is restart-idempotent: an already consumed consequence hash cannot generate a duplicate action.

Table 3: Initial continuous real-outcome integration results.

Measure	Result
Independently revealed cycles processed	4
Verified cognitive responses	4/4
Source models retained	4
External-change cycles	1
External changes incorrectly sent to self-repair	0
Private attribution audit	4/4
Duplicate actions after restart	0
Unsafe live writes	0

The four authorities were CPython, Node.js, PyPI NumPy, and Open-Meteo Madrid. The initial live cohort detected genuine Node.js and Madrid weather changes. CAU promoted both procedures:

- `procedure_real_outcome_failure_arbitration_v1`; and
- `procedure_continuous_real_outcome_learning_v1`.

The outcome-learning service runs beside the canonical runtime under heartbeat supervision.

1.4 Verified Cross-Domain Native Routing

AION reconstructed 55 provenance-bearing goal–action–outcome trajectories across five cognitive surfaces: mission language, causal strategy, private repair, tool acquisition, and changing public

outcomes. Three private neural heads learned to propose which specialist should receive a new goal. For semantic representation $x \in \mathbb{R}^{384}$ and private heads h_j , the routing proposal is

$$\hat{p}(s \mid x) = \frac{1}{3} \sum_{j=1}^3 \text{softmax}(h_j(x)), \quad s \in \{1, \dots, 5\}. \quad (5)$$

Low-confidence or low-margin predictions abstain. Downstream HexCore verification or the complete symbolic fallback remains mandatory.

Table 4: Cross-domain native-router transfer results.

Measure	Result
Verified trajectories	55
Cognitive domains	5
Independent private heads	3
Source-disjoint evaluation trajectories	23
Top-choice accuracy	95.65%
Weakest-domain accuracy	83.33%
Final verified success with fallback	100%
Specialist-evaluation reduction	64.71%
Learned parameters	37,455
Unsafe neural acceptances	0
OOD abstention	Passed

The promoted router is `procedure_verified_cross_domain_native_router_v1`. Its weights are replaceable; removing them restores the complete non-neural route.

1.5 Continual Multi-Skill Cognitive Composition

The next stage addressed broad objectives requiring more than one specialist. A monolithic continuation network was rejected after new language interfered with protected routing. The retained architecture freezes the established atomic router and gives broader natural-language clauses separate private composition capacity. A mission is represented as a directed acyclic graph

$$G_M = (V_M, E_M), \quad (6)$$

where every vertex is a proposal-only specialist contract and every edge is a verified dependency. Each vertex is independently verified or falls back; failure of one local proposal does not discard unrelated completed work.

The promoted procedure is `procedure_continual_native_cognitive_composition_v2`. The five specialist families and clause segmentation remain engineered. The next boundary is open decomposition: inventing a new specialist contract and dependency structure when no retained vocabulary expresses the goal.

1.6 Adaptive Pre-Action Metacognition

AION now performs an independent non-LLM review between planning and action commitment. The review returns one of

$$d_{\text{pre}} \in \{\text{execute, revise, investigate, abstain, escalate}\}. \quad (7)$$

Table 5: Continual native cognitive-composition results.

Measure	Result
Protected parent trajectories	55
Protected parent transfer	95.65%
New verified curriculum clauses	25
Sealed clause routing	100%
Exact multi-skill missions	5/5
Monolithic single-specialist control	0/5
Specialist-evaluation reduction	68.57%
Unsafe neural acceptances	0

Every action receives cheap executable, consent, and contradiction checks. Deep review is activated only by elevated risk, uncertainty, irreversibility, known contradiction, a counterfactual challenge, or relevant failure history. It then audits assumptions, simulates an opposing response, recalls matched failures, checks rollback, and compares alternatives.

The metacognitive pressure used for routing review depth is

$$P_{\text{meta}} = \min(1, 0.35r + 0.35u + 0.20h + 0.10i), \quad (8)$$

where r is declared risk, u is uncertainty, h is matched historical failure rate, and $i \in \{0, 1\}$ denotes irreversibility. Failure memory is indexed by action type, risk tier, irreversibility, and rollback availability; it is not applied to an entire domain.

The first challenger prevented every trap but over-deliberated on all later actions in the same domain. It was rejected. Signature-scoped failure memory removed this generalisation error.

Table 6: Adaptive pre-action metacognition results.

Measure	Result
Actor-only success	50%
Metacognitive success	100%
Weakest-domain success	100%
Tempting traps prevented	5/5
Routine actions retained on cheap path	5/5
Deliberation reduction versus always-deep review	31.25%
Unsafe executions	0
LLM calls	0
Restart retention	Passed

The sealed decisions covered chess, software, research, causal inference, and tool use. CAU promoted `procedure_adaptive_metacognitive_deliberation_v1`.

1.7 Post-Action Reflection and Scoped Lessons

Pre-action criticism prevents foreseeable mistakes; it cannot learn from an outcome that has not occurred. A second metacognitive stage therefore runs after observation and ordinary failure criticism. It compares predicted success $\hat{s}$ with observed score s ,

$$e_{\text{pred}} = |\hat{s} - s|, \quad (9)$$

and activates reflection when an outcome is bad or mediocre, prediction error is material, or the original decision was high risk. An unsurprising routine success generates a cheap no-op.

The reflector distinguishes six causes:

$$c \in \{\text{assumption, decision model, execution, evidence, environment, authority}\}. \quad (10)$$

This distinction prevents blame leakage. A broken execution adapter does not prove that the strategy was wrong; environmental change does not prove that an earlier model was always false; and weak evidence does not by itself authorize a world-model rewrite.

A retained lesson is the tuple

$$\ell = (\sigma, c, Q, a, \gamma, n, H), \quad (11)$$

where σ is the matching decision signature, c is attribution, Q is the set of required future checks, a is the recommended adjustment, γ is confidence, n is support count, and H is the evidence hash. New lessons are provisional and scoped to σ . A single trip therefore does not teach the global rule “walking is dangerous”; it can teach that a matching movement decision under an unverified clear-path assumption requires an obstacle or counterexample check.

Table 7: Post-action reflective-metacognition results.

Measure	Result
Failure-attribution families	5
Attribution accuracy	100%
Matching lesson transfer	100%
Unrelated cheap-action preservation	100%
Routine expected-success reflection	Cheap no-op
Unsafe executions	0
LLM calls	0
Restart relearning	0

The five sealed families tested assumption, execution, environment, evidence, and authority failure. Every lesson affected the matching future action while an unrelated read-only action remained on the cheap path. CAU promoted `procedure_outcome_reflective_metacognition_v2`.

1.8 Integrated Runtime Status

The canonical cognitive runtime, continuous outcome-learning service, and long-duration Arena v15 campaign operate concurrently. At the documented campaign snapshot, four of the required twelve cycles had completed over approximately 2.03 hours. Four independent remote authorities were reachable; every completed execution and transaction passed; and real Node.js and Madrid-weather changes had been detected. The campaign correctly remained unpromoted because its contract requires both twelve cycles and 24 elapsed hours.

The verified trajectory corpus was subsequently refreshed after promotion of post-action reflection: Negative artifacts remain criticism, abstention, and calibration evidence; they are not treated as positive training authority.

1.9 Capability Boundary and Next Gate

This sequence establishes a restart-persistent, governed chain from natural mission through specialist composition, action, consequence, reflective learning, private repair, and retained improvement.

Table 8: Current verified-trajectory corpus summary.

Measure	Result
Committed artifacts	138
Verified positive artifacts	105
Verified negative artifacts	21
Protocol-only artifacts	2
Unresolved artifacts	10
Capability families	7
Procedure split leakage	0

It is not evidence of subjective consciousness, unrestricted language understanding, unrestricted self-modification, or AGI. Failure categories, specialist families, semantic prototypes, benchmark counterfactuals, and several outcome authorities remain engineered.

The next decisive gate is *open diagnostic reflection*. When a real outcome is poor but no authority supplies a clean failure label, AION must:

1. preserve competing explanations for decision, assumption, evidence, execution, and environmental failure;
2. invent the minimum-cost diagnostic observation or intervention that distinguishes those explanations;
3. commit its prediction before receiving the diagnostic consequence;
4. update only the component supported by the revealed evidence;
5. test the resulting lesson on a later source-disjoint consequence; and
6. retain or retire the lesson without contaminating unrelated actions.

This joins metacognition to active causal investigation rather than relying on a development-supplied explanation of why an action failed.